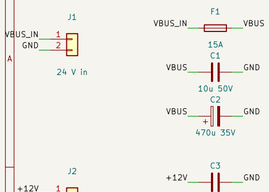
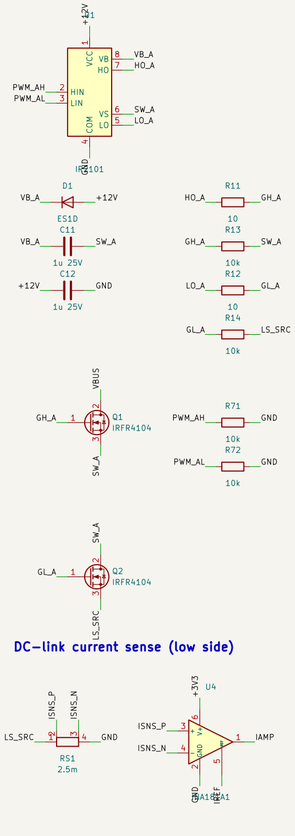
Power input, bus capacitors



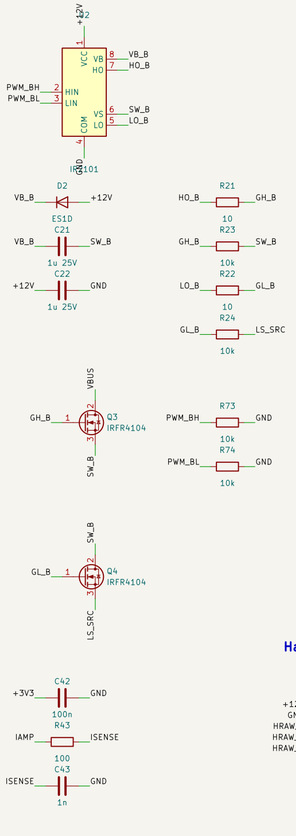
Power flags (ERC)



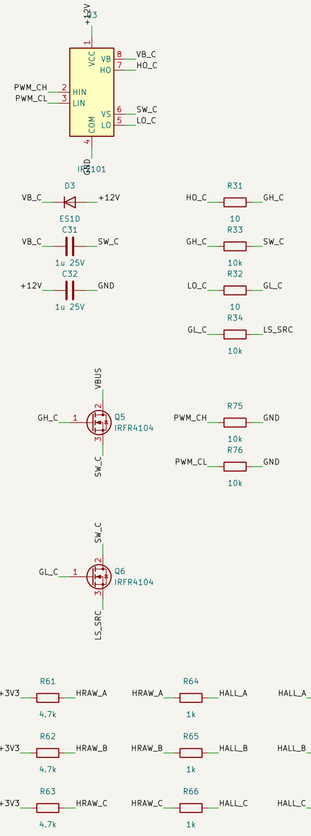
Leg A: IR2101 + half-bridge



Leg B: IR2101 + half-bridge



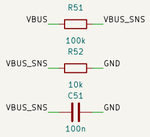
Leg C: IR2101 + half-bridge



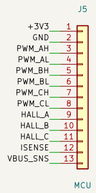
DC-link current sense (low side)

Hall sensor inputs

Bus voltage sense



MCU header



Simulated design only: nothing built in hardware.
Six-step (trapezoidal) drive: in each 60 deg sector one high side switches at 20 kHz with its low side as synchronous rectifier, one low side is on, one leg floats. Dead time (500 ns) is inserted by the firmware (compare pair per leg, src/bldc/bldc.c): the IR2101 has no internal interlock, so the MCU timing is the only protection against shoot-through (sim/ngspice/leg_deadtime.cir: shoot-through below about 250 ns).
Gate pull-downs (R13/R14,..) hold the bridge off while the MCU is in reset; R71-R76 hold the driver inputs low.
RS1 2.5 mOhm + U4 INA181A1 (gain 20): ISENSE = 1.65 V + 50 mV/A, sampled at the PWM counter peak.
VBUS_SNS = VBUS / 11. Hall sensors: open collector, 12 V supply, 4.7k pull-ups to 3.3 V, 1k / 1 nF filter.
J5 goes to the MCU: PWM_xH / PWM_xL from a centre-aligned timer, HALL_A-C to GPIO, ISENSE / VBUS_SNS to ADC.

3 x IR2101, 6 x IRFR4104, 2.5 mOhm shunt + INA181A1, Hall inputs, MCU header
Simulated design only: nothing built in hardware

Shon Zinger

Sheet: /
File: bldc.kicad_sch

Title: 05 — BLDC 6-step controller: inverter and gate drive (simulated)

Size: A2	Date: 2026-11-03	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 1/1